

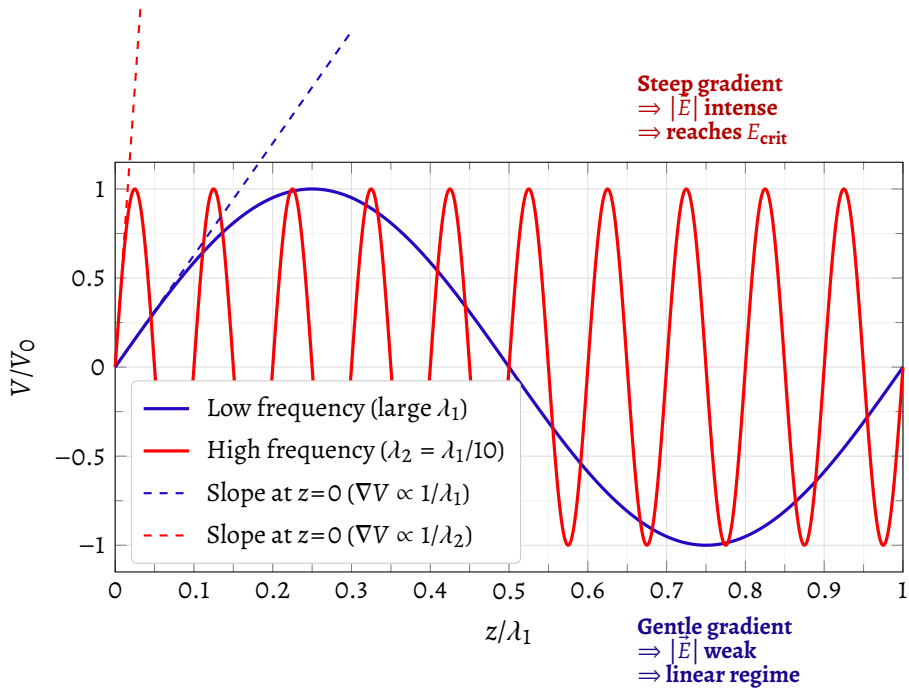


Figure 7 – Comparison of the electric potential $V(z)$ for two waves with the same amplitude V_0 , but with different frequencies, at the instant $t = 0$. The high-frequency wave (red, $\lambda_2 = \lambda_1/10$) exhibits a spatial gradient **10 times steeper** than the low-frequency wave (blue). By the relation $\vec{E} = -\nabla V$, this results in an electric field 10 times more intense. It is precisely this effect that causes the high-frequency modes to reach E_{crit} first, activating the nonlinear suppression.

